

# Linear Models Lecture 12: Hypothesis Testing

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April 29, 2026

## Where We Are

**Last lecture:** We derived the Wald statistic

$$W = (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r})' [\mathbf{R}\hat{\mathbf{V}}\mathbf{R}']^{-1} (\mathbf{R}\hat{\boldsymbol{\beta}} - \mathbf{r}) \xrightarrow{d} \chi_q^2$$

and showed how to construct robust tests and confidence intervals.

**Today:** We go deeper into the *practice* of hypothesis testing.

- Constrained least squares: equality *and* inequality constraints
- Partial identification: when the data only pins  $\boldsymbol{\beta}$  to a feasible set
- The F test (recap), Score (LM) tests, and the trinity of classical tests
- Test inversion for confidence regions
- Multiple testing and Bonferroni corrections
- Power: what determines whether you can detect an effect?
- Hansen's practical advice for applied work

## Constrained Least Squares: Setup

Many tests today compare an **unrestricted** model to a **restricted** one. We need to know how to estimate under restrictions.

**Problem:** Minimize the sum of squared errors subject to  $q$  linear constraints:

$$\tilde{\beta}_{\text{CLS}} = \arg \min_{\beta} \sum_{i=1}^n (Y_i - \mathbf{X}_i' \beta)^2 \quad \text{subject to} \quad \mathbf{R}' \beta = \mathbf{r}$$

where  $\mathbf{R}$  is  $k \times q$  and  $\mathbf{r}$  is  $q \times 1$ .

### Examples:

- $\beta_3 = 0$ : set  $\mathbf{R}' = (0, 0, 1, 0, \dots)$ ,  $\mathbf{r} = 0$  (exclusion restriction)
- $\beta_2 = \beta_3$ : set  $\mathbf{R}' = (0, 1, -1, 0, \dots)$ ,  $\mathbf{r} = 0$  (equality restriction)
- $\beta_2 + \beta_3 = 1$ : set  $\mathbf{R}' = (0, 1, 1, 0, \dots)$ ,  $\mathbf{r} = 1$  (adding-up constraint)

## The CLS Estimator

Solve via Lagrange multipliers. The solution is:

$$\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}} - (\mathbf{X}'\mathbf{X})^{-1}\mathbf{R} [\mathbf{R}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}]^{-1} (\mathbf{R}'\hat{\beta}_{\text{OLS}} - \mathbf{r})$$

### Reading the formula:

- Start from unrestricted OLS  $\hat{\beta}_{\text{OLS}}$ , subtract a correction toward the constraint
- Correction is proportional to  $(\mathbf{R}'\hat{\beta}_{\text{OLS}} - \mathbf{r})$ : how far OLS violates  $H_0$
- If OLS already satisfies the restriction,  $\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}}$

The restricted residuals and variance estimate are:

$$\tilde{e}_i = Y_i - \mathbf{x}_i' \tilde{\beta}_{\text{CLS}}, \quad \tilde{\sigma}^2 = \frac{1}{n} \sum_{i=1}^n \tilde{e}_i^2$$

Since the restriction constrains the parameter space:  $\tilde{\sigma}^2 \geq \hat{\sigma}^2$  always.

## CLS in Practice

**Simple case:** If  $H_0$  sets some coefficients to zero, CLS just means dropping those regressors.

```
# Unrestricted
mod_U <- lm(lwage ~ education + exper + I(exper^2) +
            female + union, data = wages)
# Restricted (H0: beta_female = beta_union = 0)
mod_R <- lm(lwage ~ education + exper + I(exper^2),
            data = wages)
```

**General case:** For arbitrary linear restrictions (e.g.,  $\beta_2 = \beta_3$ ), reparameterize or use the closed-form formula.

The key outputs from CLS are  $SSE_R = \sum \tilde{e}_i^2$  and  $\tilde{\sigma}^2$ , which we use to build the F and Score statistics.

## Worked Example: CLS for a Linear Restriction

**Setup.** Wage regression with separate returns to general experience and firm-specific tenure:

$$\log w_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 \text{exper}_i + \beta_3 \text{tenure}_i + e_i.$$

Becker–Mincer human-capital theory predicts these have *different* returns: firm-specific human capital should pay more at the current employer than general experience does. We want to *test* this — formally,  $H_0 : \beta_2 = \beta_3$  vs.  $H_1 : \beta_2 \neq \beta_3$ .

**Important framing.** We do not impose the restriction because we believe it. We compute the restricted estimator to obtain  $\text{SSE}_R$ , which feeds the  $F$  statistic that *tests* the restriction.

## Worked Example: Two Ways to Compute CLS

**Approach 1: Reparameterize.** Under  $H_0$  the model becomes

$$\log w_i = \beta_0 + \beta_1 \text{educ}_i + \beta_2 (\text{exper}_i + \text{tenure}_i) + e_i,$$

which is just OLS on the new combined regressor:

```
mod_R <- lm(lwage ~ educ + I(exper + tenure), data = wages)
```

**Approach 2: Apply the boxed CLS formula directly** with  $\mathbf{R}' = (0, 0, 1, -1)$  and  $\mathbf{r} = 0$ .

Both give the same  $\tilde{\beta}_{\text{CLS}}$  and the same  $\text{SSE}_R = \sum \tilde{e}_i^2$ , which feed the  $F$  statistic developed later in this lecture.

## Beyond Equality: Inequality Constraints

The CLS formulation extends naturally to inequalities:

$$\tilde{\beta}_{\text{CLS}} = \arg \min_{\beta \in \mathcal{B}} \sum_{i=1}^n (Y_i - \mathbf{x}_i' \beta)^2$$

where the **feasible set**  $\mathcal{B} \subseteq \mathbb{R}^k$  is defined by linear inequalities  $\mathbf{R}'\beta \leq \mathbf{r}$ .

### Examples:

- **Sign restrictions:**  $\beta_j \geq 0$  (a variance, a price-elasticity sign, a share)
- **Box constraints:**  $0 \leq \beta_j \leq 1$  (mixing weights, probabilities)
- **Monotonicity:**  $\beta_1 \leq \beta_2 \leq \beta_3$  (ordered effects)
- **Theory bounds:**  $\beta \in [\underline{\beta}, \overline{\beta}]$  from prior knowledge

The feasible set  $\mathcal{B}$  is a *polyhedron* (an intersection of half-spaces) rather than a single point or a linear subspace.



## Solving Inequality CLS: Two Cases

**Strategy.** Solve unconstrained OLS first, then check whether the result satisfies the constraints.

**Case 1:**  $\hat{\beta}_{\text{OLS}} \in \mathcal{B}$ . No constraint binds; the constrained and unconstrained problems have the same solution:

$$\tilde{\beta}_{\text{CLS}} = \hat{\beta}_{\text{OLS}}.$$

**Case 2:**  $\hat{\beta}_{\text{OLS}} \notin \mathcal{B}$ . The constrained solution sits on the boundary of  $\mathcal{B}$ . On the face where one or more constraints bind, those constraints hold with equality, and the solution is given by the *equality*-CLS formula from earlier in this lecture, applied to the binding constraints only.

## Inequality CLS as a Projection

Conceptually, you are projecting  $\hat{\beta}_{OLS}$  onto the feasible region  $\mathcal{B}$ .

Picking which constraints bind is what makes the inequality case harder than the equality case — in practice this is solved by a *quadratic program*. In R: `quadprog::solve.QP()`.

## Why Inequality Constraints? Partial Identification

Sometimes the data and minimal assumptions only tell us  $\beta$  lies in a *set*, not a point.

**The identified set:**  $\Theta_I = \{\beta : \beta \text{ is consistent with the data and assumptions}\}.$

- **Point identification:**  $\Theta_I = \{\beta_0\}$  — a single value
- **Partial identification:**  $\Theta_I$  is a non-degenerate set, often a polyhedron

**Where it shows up:** missing data, censoring, interval-valued outcomes, treatment effect bounds, set-identified moment inequalities.

The identified set *is* a feasible set  $\mathcal{B}$ . Estimation under partial identification reports both a point (e.g., the inequality-CLS solution on  $\mathcal{B}$ ) and the *width* of  $\mathcal{B}$  — and the width does *not* shrink with  $n$ .

## A Worked Example: NSW Job Training

The **National Supported Work Demonstration** (1976–77) randomly assigned disadvantaged workers to a 9–12 month training program ( $X_i = 1$ ) or a control group ( $X_i = 0$ ). LaLonde (1986) used it to benchmark non-experimental evaluators.

**Outcome.**  $Y_i = 1$  if subject  $i$  had positive earnings in 1978 (employed at follow-up).

**Sample** (Dehejia–Wahba experimental sample,  $n = 445$ ):

|                       | $n$ | Employed in 1978 | $\bar{Y}_{\text{obs}}$ |
|-----------------------|-----|------------------|------------------------|
| Treatment ( $X = 1$ ) | 185 | 140              | 0.757                  |
| Control ( $X = 0$ )   | 260 | 168              | 0.646                  |

## NSW: Naive ATE and a Missing-Data Thought Experiment

**Naive ATE:**  $\bar{Y}_{\text{obs}}(1) - \bar{Y}_{\text{obs}}(0) = 0.757 - 0.646 = 0.111$  (an 11 pp employment effect).

NSW's actual follow-up was nearly complete. **What if 30% had been lost?** The next slides treat that as a thought experiment to see what the data could still pin down.

## The Manski Decomposition (30% Missing)

Suppose only 70% of subjects could be located in 1978. Let  $D_i = 1$  if observed;  $P(D_i = 1) = 0.7$ .

For each arm, by the law of total probability:

$$E[Y \mid X = x] = \underbrace{p(x) \bar{Y}_{\text{obs}}(x)}_{\text{identified}} + \underbrace{(1 - p(x)) E[Y \mid X = x, D = 0]}_{\text{unknown}}$$

The first term is identified from observed respondents. The second is unknown — but since  $Y \in \{0, 1\}$ , the unknown conditional mean lies in  $[0, 1]$ , so:

$$p(x) \bar{Y}_{\text{obs}}(x) \leq E[Y \mid X = x] \leq p(x) \bar{Y}_{\text{obs}}(x) + (1 - p(x))$$

The lower bound replaces every missing  $Y$  with 0 (worst case for that arm); the upper bound replaces every missing  $Y$  with 1. These are **sharp**: no narrower bound is consistent with the data alone.

## Worst-Case Bounds: Apply to NSW

Plug in  $p(x) = 0.7$  for each arm.

**Treatment** ( $\bar{Y}_{\text{obs}} = 0.757$ ):

$$0.7 \cdot 0.757 \leq E[Y \mid X=1] \leq 0.7 \cdot 0.757 + 0.3 \implies E[Y \mid X=1] \in [0.530, 0.830]$$

**Control** ( $\bar{Y}_{\text{obs}} = 0.646$ ):

$$0.7 \cdot 0.646 \leq E[Y \mid X=0] \leq 0.7 \cdot 0.646 + 0.3 \implies E[Y \mid X=0] \in [0.452, 0.752]$$

**ATE bounds.**  $\theta = E[Y \mid X=1] - E[Y \mid X=0]$  is smallest when arm 1 is at its lower bound and arm 0 at its upper bound (largest in the opposite corner):

$$\theta \in [0.530 - 0.752, 0.830 - 0.452] = [-0.22, 0.38]$$

The naive estimate 0.11 sits inside, but the bound is wide enough that we cannot even sign the effect. Width = 0.60.

## Tighter Bounds via Monotone Selection

**Plausible assumption.** Subjects lost to follow-up — transient, moving, out of the labor force — are weakly *less* likely to be employed than those we observe in the same arm:

$$E[Y \mid X = x, D = 0] \leq E[Y \mid X = x, D = 1] = \bar{Y}_{\text{obs}}(x).$$

**Effect.** The upper bound now replaces missing  $Y$  with  $\bar{Y}_{\text{obs}}(x)$ , not with 1, so  $E[Y \mid X=x] \leq \bar{Y}_{\text{obs}}(x)$ . The lower bound is unchanged.

**Apply to NSW:**

$$E[Y \mid X=1] \in [0.530, 0.757], \quad E[Y \mid X=0] \in [0.452, 0.646],$$

$$\theta \in [0.530 - 0.646, 0.757 - 0.452] = [-0.12, 0.30].$$

Width drops from 0.60 to 0.42. Still crosses zero — an 11 pp underlying effect is too small for one mild assumption to settle the sign at 30% missing.



## The Identified Set as an Inequality-CLS Feasible Set

Write the CEF as a linear regression  $E[Y | X] = \beta_0 + \beta_1 X$ , so  $\beta_0 = E[Y | X=0]$  and  $\beta_0 + \beta_1 = E[Y | X=1]$ .

The four MTS bounds map directly to four linear inequalities on  $\beta = (\beta_0, \beta_1)'$ :

$$\mathcal{B} = \{\beta : 0.452 \leq \beta_0 \leq 0.646, 0.530 \leq \beta_0 + \beta_1 \leq 0.757\}.$$

**Inequality CLS** on the observed sample:

$$\tilde{\beta}_{\text{CLS}} = \arg \min_{\beta \in \mathcal{B}} \sum_{i: D_i=1} (Y_i - \beta_0 - \beta_1 X_i)^2.$$

$\mathcal{B}$  is the identified set: it pins  $\beta_1$  to the interval  $[-0.12, 0.30]$ . An inequality-CLS point estimate is just one  $\beta$  inside  $\mathcal{B}$ ; reporting that point hides the bound. Report the interval.

## Computing the Bounds in R

```
u <- "https://users.nber.org/~rdehejia/data/nswre74_"
cn <- c("treat", "age", "educ", "black", "hispan", "married",
        "nodegree", "re74", "re75", "re78")
d <- rbind(read.table(paste0(u, "treated.txt"), col.names=cn),
            read.table(paste0(u, "control.txt"), col.names=cn))
y1 <- mean(d$re78[d$treat==1] > 0)      # 0.7568      (n = 185)
y0 <- mean(d$re78[d$treat==0] > 0)      # 0.6462      (n = 260)
p <- 0.7                                # 70% located, 30% missing
ate_wc <- c(p*y1 - (p*y0+1-p), (p*y1+1-p) - p*y0) # [-0.22, 0.38]
ate_mts <- c(p*y1 - y0, y1 - p*y0)          # [-0.12, 0.30]
```

**What to report:** “The NSW employment ATE lies in  $[-0.12, 0.30]$  under monotone selection (naive estimate 0.11).”

## Setting Up the Linear Program

Reparameterize from  $\beta$  to the conditional means  $\mu = (\mu_0, \mu_1)'$  where  $\mu_x = E[Y | X=x]$ . Then  $\beta_0 = \mu_0$  and  $\beta_1 = \mu_1 - \mu_0 = \mathbf{c}'\mu$  with  $\mathbf{c} = (-1, 1)'$ .

Each MTS bound is a row  $\mathbf{a}_j'\mu \{ \geq, \leq \} r_j$ . Stack the four:

$$\underbrace{\begin{pmatrix} 1 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 1 \end{pmatrix}}_A \underbrace{\begin{pmatrix} \mu_0 \\ \mu_1 \end{pmatrix}}_{\text{dir}} \underbrace{\begin{matrix} \geq \\ \leq \\ \geq \\ \leq \end{matrix}}_{\text{dir}} \underbrace{\begin{pmatrix} 0.452 \\ 0.646 \\ 0.530 \\ 0.757 \end{pmatrix}}_{\text{rhs}}.$$

Why  $\mu$  and not  $\beta$ ? lpSolve defaults variables to  $\geq 0$ ; conditional means satisfy this automatically, but  $\beta_1$  can be negative.

## Bounds via lpSolve

Two LPs — minimize and maximize  $\mathbf{c}'\boldsymbol{\mu} = \mu_1 - \mu_0$  over the same feasible set:

```
library(lpSolve)
A      <- rbind(c(1,0), c(1,0), c(0,1), c(0,1))
dir    <- c(">=", "<=", ">=", "<=")
rhs    <- c(0.452, 0.646, 0.530, 0.757)
c(b1_lo = lp("min", c(-1,1), A, dir, rhs)$objval,
  b1_hi = lp("max", c(-1,1), A, dir, rhs)$objval)
# b1_lo  b1_hi:  -0.116  0.305
```

lpSolve confirms  $\beta_1 \in [-0.12, 0.30]$ . More parameters or constraints just adds rows to  $A$ ,  $dir$ , and  $rhs$ .

## The F Test (Recap from Section)

Compare the unrestricted SSE to the restricted (CLS) SSE. Testing  $H_0: \mathbf{R}'\boldsymbol{\beta} = \mathbf{r}$  with  $q$  restrictions:

$$F = \frac{(\text{SSE}_R - \text{SSE}_U)/q}{\text{SSE}_U/(n-k)} = \frac{W^0}{q}$$

- **Distribution:** under normal homoskedastic errors,  $F \sim F_{q, n-k}$  exactly; asymptotically (any iid errors),  $qF \xrightarrow{d} \chi_q^2$ .
- **Heteroskedasticity:** the  $F_{q, n-k}$  distribution is no longer valid — use the robust Wald test from Lecture 11.
- **In R:** `anova(mod_R, mod_U)`.

The package-default “F-statistic that all slopes equal zero” is rarely a scientifically interesting hypothesis. Hansen: there is no reason to report it.

## Score Tests: The Idea

The Wald test starts from the **unrestricted** estimate and asks: is  $\hat{\theta}$  far from  $\theta_0$ ?

The **Score test** (also called **Lagrange Multiplier test**) starts from the **restricted** estimate and asks: does the objective function want to move away from the restriction?

- Compute the restricted estimator  $\tilde{\beta}$  (constrained least squares)
- Evaluate the *gradient* (score) of the objective function at  $\tilde{\beta}$
- If  $H_0$  is true, the score should be near zero (we're near the optimum)
- If  $H_0$  is false, the score should be large (the restriction is pulling us away)

## Score Statistic for Linear Restrictions

For  $H_0: \mathbf{R}'\boldsymbol{\beta} = \mathbf{c}$  in the normal regression model:

$$S = \frac{(\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c})'[\mathbf{R}'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{R}]^{-1}(\mathbf{R}'\hat{\boldsymbol{\beta}} - \mathbf{c})}{\tilde{\sigma}^2}$$

- Identical to  $W^0$  except  $s^2$  replaced by  $\tilde{\sigma}^2$  (restricted variance).
- Under  $H_0$ :  $S \xrightarrow{d} \chi_q^2$

**Key result:**  $S$  is a monotone transformation of  $F$ :  $S = n \left( 1 - \frac{1}{1 + qF/(n-k)} \right)$ , so the Score test and the F test always give the same accept/reject decision.

## When Are Score Tests Useful?

For linear regression with linear restrictions, the Score test offers nothing new over  $F$  or Wald.

**But** in more complex settings, Score tests have a major advantage:

- They only require estimation under  $H_0$  (the restricted model)
- The “unrestricted” model may not even be a simple OLS regression

**Example:** Testing for heteroskedasticity (Breusch–Pagan). The null is homoskedastic OLS—easy. The unrestricted model allows  $\text{Var}(e_i | X_i)$  to vary with  $X$ , which requires modeling the variance function. The Score test only needs the OLS residuals.



## Score Tests: More Settings Where They Help

The same logic — estimate only under  $H_0$ , evaluate the score — applies whenever the null gives a clean model and the alternative is messy.

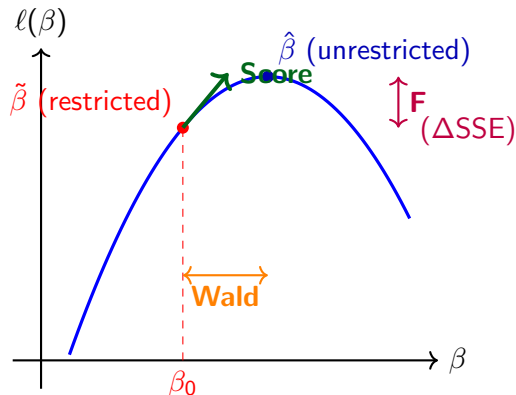
- **Serial correlation** (Breusch–Godfrey): null is OLS with iid errors.
- **Omitted nonlinearities**: null is a simple linear specification.
- Any setting where  $H_0$  gives you a clean model but  $H_1$  is messy.

## Three Ways to Test the Same Hypothesis

|                                | Wald                                                          | Score (LM)                               | F / LR-like                    |
|--------------------------------|---------------------------------------------------------------|------------------------------------------|--------------------------------|
| <b>Estimates from Measures</b> | Unrestricted<br>Distance of $\hat{\theta}$<br>from $\theta_0$ | Restricted<br>Gradient at<br>restriction | Both<br>Change in<br>fit (SSE) |
| <b>Null dist.</b>              | $\chi_q^2$                                                    | $\chi_q^2$                               | $\chi_q^2/q$ or $F_{q,n-k}$    |
| <b>Robust?</b>                 | Yes (sandwich)                                                | Score-like                               | No                             |

For linear restrictions in the normal model, all three are monotone transformations of each other and give identical decisions. They differ for nonlinear restrictions or non-normal models.

# Visualizing the Trinity



- **Wald**: horizontal distance between  $\hat{\beta}$  and  $\beta_0$
- **Score**: slope at restricted estimate
- **F**: vertical distance ( $\Delta\text{SSE}$ )

## Confidence Intervals Are Inverted Tests

Recall the standard 95% CI:

$$\hat{C} = [\hat{\theta} - 1.96 \cdot s(\hat{\theta}), \hat{\theta} + 1.96 \cdot s(\hat{\theta})]$$

This is exactly the set of values  $\theta$  that are *not rejected* by a two-sided  $t$ -test:

$$\hat{C} = \{\theta : |T(\theta)| \leq 1.96\}$$

**General principle:** Inverting a test with good Type I error control produces a confidence set with good coverage.

$$P[\theta \in \hat{C}] = P[\text{Accept} \mid \theta] = 1 - P[\text{Type I error}]$$

## Why Test Inversion Matters: Nonlinear Parameters

Consider  $\theta = \beta_1/\beta_2$  (e.g., peak experience in a wage equation).

**Approach 1:** Delta method CI

$$\hat{\theta} \pm 1.96 \cdot s(\hat{\theta})$$

From Lecture 11, this can be inaccurate for ratios (Fieller's problem).

**Approach 2:** Rewrite as a *linear* restriction  $\beta_1 - \theta\beta_2 = 0$  and invert:

$$\hat{C} = \left\{ \theta : \frac{(\hat{\beta}_1 - \theta\hat{\beta}_2)^2}{\mathbf{R}'(\theta)\hat{\mathbf{V}}\mathbf{R}(\theta)} \leq 1.96^2 \right\}$$

where  $\mathbf{R}(\theta) = (1, -\theta)'$ . This requires a grid search over  $\theta$ .

## Test Inversion vs. Delta Method: Hansen's Example

**Setup.** Peak experience in a wage equation,  $\theta = -50\beta_1/\beta_2$ .

- **Delta method CI:** [29.8, 29.9]
- **Inverted linear-test CI:** [29.1, 30.6]

The delta method interval is **far too narrow** — the ratio's sampling distribution is poorly approximated by a normal centered at  $\hat{\theta}$ . Inverting the test for the equivalent linear restriction does not require that approximation.

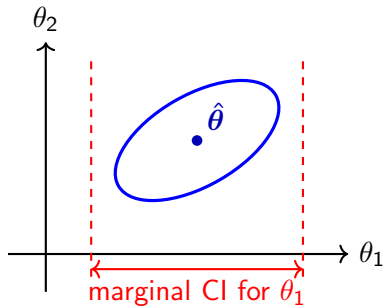
## Test Inversion in R

```
# Wage equation: log(wage) ~ exper + exper^2/100 + ...
mod <- lm(lwage ~ exper + I(exper^2/100) + education,
          data = wages)
b <- coef(mod); V <- vcovHC(mod)

# Grid search: invert the t-test for theta = -50*b1/b2
theta_grid <- seq(20, 40, by = 0.01)
in_CI <- sapply(theta_grid, function(th) {
  R <- c(0, 1, -th/50, 0) # gradient of b[2] - th*b[3]/50
  num <- (b[2] - th * b[3] / 50)^2
  den <- t(R) %*% V %*% R
  num / den <= qchisq(0.95, 1)
})
CI <- range(theta_grid[in_CI])
```

## Confidence Regions for Multiple Parameters

For  $q > 1$  parameters, invert the Wald test:  $\hat{C} = \{\boldsymbol{\theta} : W(\boldsymbol{\theta}) \leq \chi_{q, 1-\alpha}^2\}$  — an **ellipsoid** in  $\mathbb{R}^q$  centered at  $\hat{\boldsymbol{\theta}}$ , shaped by  $\hat{\mathbf{V}}_{\boldsymbol{\theta}}^{-1}$ .



Marginal CIs (projections) are always wider than the ellipsoid in each dimension.



## The Multiple Testing Problem

Suppose you test  $k$  hypotheses, each at level  $\alpha = 0.05$ . Under the **global null** (all  $k$  true), what is  $P[\text{at least one false rejection}]$ ?

By **Boole's inequality**:

$$P\left[\min_{j \leq k} p_j < \alpha\right] \leq \sum_{j=1}^k P[p_j < \alpha] \rightarrow k\alpha$$

| $k$ (tests) | $\alpha$ per test | Familywise error $\leq$ |
|-------------|-------------------|-------------------------|
| 5           | 0.05              | 0.25                    |
| 10          | 0.05              | 0.50                    |
| 20          | 0.05              | 1.00                    |

With 20 tests, you are *virtually certain* to get a false rejection!

## The Bonferroni Correction

**Goal:** Control the *familywise error rate* (FWER) at  $\alpha$ .

**Rule:** Reject the  $j$ th hypothesis only if  $p_j < \alpha/k$ . Equivalently:  $p_{\text{Bonf}} = k \cdot \min_{j \leq k} p_j$ .

**Proof:**

$$P\left[\min_{j \leq k} p_j < \frac{\alpha}{k}\right] \leq \sum_{j=1}^k P\left[p_j < \frac{\alpha}{k}\right] \rightarrow k \cdot \frac{\alpha}{k} = \alpha$$

**Simple and conservative.** Controls FWER for *any* dependence structure among the tests. The cost: reduced power when  $k$  is large.

## Bonferroni: Worked Example

Two coefficients tested at  $\alpha = 0.05$ :

|                  | Individual $p$ | Bonferroni $p (= 2 \times p)$ |
|------------------|----------------|-------------------------------|
| Union membership | 0.04           | 0.08                          |
| Married status   | 0.15           | 0.30                          |

- **Without correction:** Union membership “significant” at 5%.
- **With Bonferroni:** Neither significant at  $\text{FWER} = 0.05$  (need  $p < 0.025$ ).

**When to worry:** exploring many specifications/subgroups, examining many coefficients, or reporting the “most significant” result.

## Bonferroni in R

```
# Get p-values from a regression
mod <- lm(lwage ~ education + exper + I(exper^2) +
          female + union + married, data = wages)
pvals <- summary(mod)$coefficients[-1, 4] # drop intercept

# Bonferroni correction
p_bonf <- p.adjust(pvals, method = "bonferroni")
cbind(raw = round(pvals, 4), bonferroni = round(p_bonf, 4))

# Other options: Holm (less conservative, still controls FWER)
p_holm <- p.adjust(pvals, method = "holm")
```

**Holm's method** is uniformly more powerful than Bonferroni while still controlling FWER. Use it when available.

## Type I and Type II Errors

|              | $H_0$ true                | $H_0$ false                    |
|--------------|---------------------------|--------------------------------|
| Reject $H_0$ | Type I error ( $\alpha$ ) | Correct (Power = $1 - \beta$ ) |
| Accept $H_0$ | Correct ( $1 - \alpha$ )  | Type II error ( $\beta$ )      |

- **Size** ( $\alpha$ ): probability of falsely rejecting a true null
- **Power** ( $\pi$ ): probability of correctly rejecting a false null:  $\pi(\theta) = P[\text{Reject } H_0 \mid \theta \neq \theta_0]$
- Power depends on the *true value* of  $\theta$ , on  $n$ , and on  $\alpha$ .

**Key trade-off:** Making  $\alpha$  smaller reduces Type I errors but also reduces power. A test with  $\alpha = 0$  never rejects and has zero power.

## Power of the $t$ -Test

For  $H_0: \theta = \theta_0$  vs.  $H_1: \theta > \theta_0$ :

$$T = \frac{\hat{\theta} - \theta_0}{s(\hat{\theta})} \approx Z + \delta, \quad Z \sim N(0, 1)$$

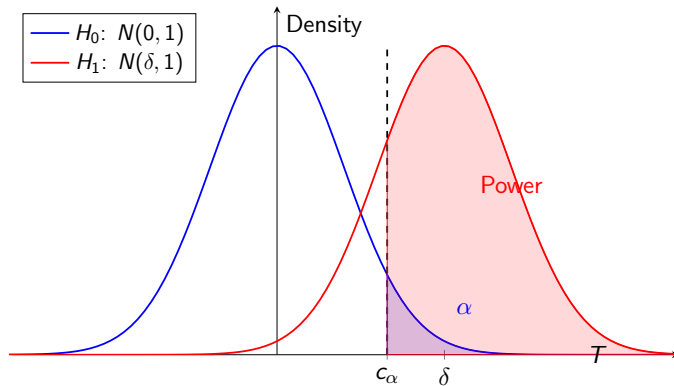
where  $\delta = (\theta - \theta_0)/s(\hat{\theta})$  is the **signal-to-noise ratio**.

Power of a one-sided test at level  $\alpha$ :  $\pi(\delta) = \Phi(\delta - z_\alpha)$

**Power increases when:**

- The true effect  $|\theta - \theta_0|$  is larger
- The standard error  $s(\hat{\theta})$  is smaller (more precise estimation)
- The significance level  $\alpha$  is larger (less stringent threshold)
- The sample size  $n$  is larger (since  $s(\hat{\theta}) \propto 1/\sqrt{n}$ )

# Visualizing Power



## Power in OLS: What You Can Control

In OLS,  $s(\hat{\theta}) \approx \sigma_e / (\text{sd}(X) \cdot \sqrt{n})$  for a single regressor. So:

$$\delta \approx \frac{(\theta - \theta_0) \cdot \text{sd}(X) \cdot \sqrt{n}}{\sigma_e}$$

### Levers for increasing power:

- 1 **Increase**  $n$ : power grows with  $\sqrt{n}$
- 2 **Reduce**  $\sigma_e$ : add controls that explain  $Y$  (reduce residual variance)
- 3 **Increase**  $\text{sd}(X)$ : more variation in the regressor of interest
- 4 **Increase**  $\alpha$ : use 5% instead of 1% (but more Type I errors)

**Controls help power!** Relevant covariates reduce  $\sigma_e$  without reducing  $\text{sd}(X)$ —a free power boost. But irrelevant covariates cost degrees of freedom.



## Power and Test Dimension

For joint tests ( $q > 1$ ), the Wald statistic under local alternatives:  $W \xrightarrow{d} \chi_q^2(\lambda)$ , where  $\lambda = \mathbf{h}' \mathbf{V}_\theta^{-1} \mathbf{h}$  is the **non-centrality parameter**.

**Critical fact:** For a fixed  $\lambda$ , power *decreases* as  $q$  increases.

| $q$ (restrictions) | $\lambda$ for 50% power | Sample size increase |
|--------------------|-------------------------|----------------------|
| 1                  | 3.85                    | baseline             |
| 2                  | 4.96                    | +28%                 |
| 3                  | 5.77                    | +50%                 |

**Takeaway:** Testing more restrictions simultaneously dilutes power. A single-coefficient  $t$ -test is more powerful than a joint  $F$ -test that includes other restrictions.

## The 50% Power Benchmark

How far must the true parameter be from  $\theta_0$  for 50% power?

### One-sided test:

- At  $\alpha = 0.05$ : need  $\delta \geq 1.65$  standard errors
- At  $\alpha = 0.01$ : need  $\delta \geq 2.33$  standard errors

**Implication for sample size:**  $(2.33/1.65)^2 \approx 2$ .

A test at  $\alpha = 0.01$  requires roughly **twice the sample size** as  $\alpha = 0.05$  to achieve the same power.

**Two-sided** ( $\alpha = 0.05$ ): need  $|\delta| \geq 1.96$  for 50% power. Since  $se \propto 1/\sqrt{n}$ , you need  $n \propto (\sigma_e/(\theta - \theta_0))^2$ .

## Power Calculation in R

```
# Approximate power for a two-sided t-test in OLS
# Inputs: effect size, residual SD, regressor SD, n, alpha
power_ols <- function(effect, sigma_e, sd_x, n, alpha=0.05) {
  se <- sigma_e / (sd_x * sqrt(n))
  delta <- effect / se
  z <- qnorm(1 - alpha/2)
  power <- pnorm(delta - z) + pnorm(-delta - z)
  return(power)
}

# Example: detect beta = 0.1 with sigma_e = 1, sd(x) = 2
sapply(c(50, 100, 200, 500, 1000), function(n)
  round(power_ols(0.1, 1, 2, n), 3))
# [1] 0.080 0.117 0.198 0.463 0.803
```

With  $\beta = 0.1$ ,  $\sigma_e = 1$ ,  $\text{sd}(X) = 2$ : you need  $n \approx 1000$  for 80% power!

# Test Consistency

## Definition

A test is **consistent against fixed alternatives** if for any true  $\theta \neq \theta_0$ :  $P[\text{Reject } H_0 \mid \theta] \rightarrow 1$  as  $n \rightarrow \infty$ .

**Good news:** The  $t$ -test and Wald test are consistent. As  $n \rightarrow \infty$ ,  $s(\hat{\theta}) \rightarrow 0$ , so  $|T| \rightarrow \infty$  whenever  $\theta \neq \theta_0$ .

**Caution:** Consistency means you will eventually reject any false null—including *economically trivial* deviations from  $\theta_0$ .

In very large samples, statistical significance  $\neq$  economic significance. A statistically significant but tiny coefficient may not be meaningful.

# Hansen's Rules for Applied Work

## 1. Report standard errors, not $t$ -ratios.

- Standard errors focus attention on precision and confidence intervals.

## 2. Report $p$ -values, not asterisks.

- $p$ -values contain more information than  $*/**/* ** *$  (“an inferior practice”).

## 3. Focus on economically motivated hypotheses.

- Don't mechanically test every coefficient against zero.
- Report the  $t$ -test for  $\beta_j = 0$  when this is a scientifically interesting question.

## More Practical Advice

### 4. “Do Not Reject” $\neq$ “Accept.”

- Failing to reject means insufficient evidence, *not* that  $H_0$  is true.
- Never write: “the regression finds that  $X$  has no effect on  $Y$ .”

### 5. Statistical significance $\neq$ economic significance.

- With large  $n$ , even tiny effects become “significant.”
- Always discuss the *magnitude* and substantive meaning.

### 6. For nonlinear hypotheses, use minimum distance or test inversion.

- The Wald statistic is *not invariant* to the algebraic formulation of  $H_0$ .
- Different equivalent formulations can give different results!

# The “What to Report” Checklist

For any coefficient of interest  $\theta$ :

- 1 **Point estimate**  $\hat{\theta}$ : the “best guess”
- 2 **Standard error**  $s(\hat{\theta})$ : measure of precision
- 3 **Confidence interval**: range of values consistent with the data
- 4 **Economic interpretation**: what the magnitude means in context

When testing:

- 5 State the **hypothesis** clearly (what question does this answer?)
- 6 Report the ***p*-value** (not just “significant” / “not significant”)
- 7 Discuss **power** if you fail to reject (could you have detected a meaningful effect?)
- 8 Account for **multiple testing** if examining many hypotheses

# Summary

| Topic            | Key Takeaway                                                                         |
|------------------|--------------------------------------------------------------------------------------|
| CLS              | Equality $R'\beta = r$ via Lagrange; gives $SSE_R$ for $F$ and Score                 |
| Inequality CLS   | Feasible set $\mathcal{B}$ as a polyhedron; project OLS onto $\mathcal{B}$ via QP    |
| Partial ID       | $\Theta_I$ has width that does not vanish with $n$ ; report bounds, not just a point |
| F test           | $F = W^0/q; \sim F_{q,n-k}$ under normal homo.; use robust Wald otherwise            |
| Score test       | From restricted model; equivalent to $F$ for linear restrictions                     |
| Trinity          | Wald, Score, $F$ agree for linear $H_0$ in normal model                              |
| Test inversion   | Confidence sets = non-rejected values; essential for nonlinear parameters            |
| Bonferroni       | With $k$ tests, use $\alpha/k$ to control familywise error                           |
| Power            | $\delta$ = effect/se; grows with $\sqrt{n}$ ; 50% power needs $\delta \approx 2$     |
| Practical advice | Report SEs, CIs, $p$ -values; focus on magnitudes; consider power                    |



# Next Time

## Lecture 13: Instrumental Variables

- What happens when  $E[\mathbf{X}\mathbf{e}] \neq \mathbf{0}$ ?
- Omitted variable bias, simultaneity, measurement error
- The IV solution: find  $\mathbf{Z}$  such that  $E[\mathbf{Z}\mathbf{e}] = \mathbf{0}$  but  $E[\mathbf{Z}\mathbf{X}'] \neq \mathbf{0}$